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Common Type System in .NET Framework

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Common Type System (CTS) in .NET Framework

In this article, I am going to discuss the **Common Type System in .NET Framework**. Please read our previous article, where we discussed the Intermediate Language (CLR)/CIL and JLS/JVM in .NET Framework and Common Type System (CTS) in .NET Framework before look into type system and declared, used and managed in the Common Language Runtime (CLR). At the end of this article, you will understand what is Common Type System (CTS) in C# and How CTS in .NET works.

What is the Common Type System in .NET Framework?

The .NET Framework supports many programming languages such as C#, VB.NET, J#, F#, etc. Every programming language has its own data type. One programming language data type cannot be understood by other programming language. But there can be situations when we need to communicate between two different programming languages. For example, we need to write code in the .NET language and that code may be called from C# language. In order to ensure smooth communication between different programming languages, Microsoft has introduced Common Type System (CTS) which ensures that data types defined in two different languages get compiled to a common data type.

CLR in .NET Framework will execute all programming languages data types. This is possible because CLR has its own data types which are common to all programming languages. At the time of compilation, all language-specific data types are converted into CTS data type. This data type system of CLR is common to all .Net Supported Programming languages and this is known as the Common Type System (CTS) in .NET Framework.

Example to understand Common Type System in .NET Framework

Let us understand Common Type System (CTS) in .NET Framework with an example. What we are going to do is, we will create two applications. One Application uses the C# Language and the other one is using VB.NET Language. And then we will try to see the C code of both of these applications and then we will try to see what the CTS looks like.

Here we are going to create two class library projects. One class library project uses the C#.NET language and the other class library project uses the VB.NET language.

Creating C# Class Library Project:

Create a class library project with the name **CsharpClassLibrary** and use the C# programming language. Once you create the C# class library project then add a class file with the name **CalculateJava** and then copy and paste the following code into it.

```
public class CalculateJava
{
    public int[] GetJavaArray()
    {
        int a = 5, b = 25;
        int[] c = a * b;
        return c;
    }
}
```

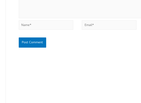
Creating VB Class Library Project:

To do same CalculateJava class, let us add another class library project with the name as **VBClassLibrary** and using **VB** as the programming language. Once you created the VB Class library project then add a class file with the name **CalculateJava** to the project and copy and paste the following code into it.

```
Public Class CalculateJava
    Public Function GetJavaArray() As Integer
        Dim a As Integer = 5
        Dim b As Integer = 10
        Dim c As Integer
        c = a * b
        Return c
    End Function
End Class
```

Now, build the solution which should generate the respective dll. In our previous article, we discussed how to use the **ILDASM** tool to view the IL code in detail. So, let us open Visual Studio Command Prompt in Administrative mode and run two instances of the **ILDASM.exe** command i.e. one for the VB class library project DLL file and the other one for the C# class library project DLL file as shown in the below image.

Now, let us open the IL code of the Calculate method of both the class library projects as shown in the below image. Please have a look at the integer variable in the IL code which is int32 in this case. In the C# class library project, we use int as the type and its decimal variable is 5, and it's stored in the IL code library project, we use integer as the data type to receive the variable is 5, and in the end of the day, both the data types are compiled to a common data type i.e. Int32.



So, whether we write the code in VB.NET or in C#.NET, it follows the dot net rules or specifications, at the end of the day it is compiled into a Common Type System (CTS) .NET Framework which is common for all .NET Supported Programming Languages. This Common Type System is Provided to CLR and is the same for all .NET Supported programming languages. While compiling the source code into the executable file, the compiler, they will convert the language-specific data type to CLR-specific data type. For example, in C# and Integer in VB is converted to Int32 after compilation and Int32 is the data type which is understood by CLR. So you can see Int32 is the CLR-specific data type.

We already demonstrated IL code and CTS in C#. So, in the next article, I am going to demonstrate to you the example of Common Language Specification (CLS) in detail. Here, in this article, I try to explain the **Common Type System (CTS) in .NET Framework** in detail and I hope you enjoy CTS (Common Type System) in the C#.NET Framework article.

Dot Net Tutorials
About the Author: Pranaya Rout
Pranaya Rout has published more than 3,000 articles in his 12 year career. Pranaya Rout has very good experience with Microsoft Technologies, including C#, ASP.NET MVC, ASP.NET Web API, ASP.NET Web MVC, EF Core, ADO.NET, LINQ, SQL Server, Azure, Angular, Bootstrap, Ajax, Jquery, Cloud Computing, Microservices, Design Patterns and still learning new technologies.

Previous Lesson
Intermediate Language (CLR)/JVM & JLS/JVM Code in C#

Next Lesson
Common Language Specification in .NET Framework

4 thoughts on "Common Type System in .NET Framework"

ANM
AUGUST 17, 2021 AT 10:20 PM
Thank you for this.
I was only able to create the IL for the VBClassLibrary. I couldn't do the same for the CsharpClassLibrary and also for the CTS program in the previous article.
Reply

AMBACHANDAN P
SEPTEMBER 20, 2021 AT 10:10 PM
Sir Ma,
I have a doubt, sir, in the IL itself the we have the common datatype (Int32). while language compiler compile source code into MSIL. But in the Base Class Library (BCL) we have common data type the CTS is a set of common data type. Why we actually need to connect to CTS, so how MSIL have a common data type for all language ? I have this doubt for sure,Sorry please help me to come out.
Reply

DOT NET TUTORIALS
MAY 9, 2022 AT 2:07 PM
The Language Compiler while generating the IL Code will follow the CTS and CLS of CLR.
Reply

MANISH
MARCH 4, 2022 AT 12:49 AM
The previous C# code is performing a simple addition of two numbers values via the Calculation method. The .NET libraries do not contain platform-specific instructions but rather use an approach, IL code that is generated using the corresponding CIL compiler (i.e.csc) during the build process.
Managed code is executed by the CLR which manages that .NET library into memory and you end up with a single .Net assembly (dll) containing a manifest, metadata, and IL instructions (instead of machine code). The Framework ships with an excellent utility to disassemble any .NET binary into its corresponding IL code referred to as "ILDASM.EXE".
Reply

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